Two collinear reflections

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Let ABC be an acute triangle. Let A_1 (resp. C_1) be the reflection of A (resp. C) in (BC) (resp. (AB)). Show that if A_1, B and C_1 are collinear and $C_1B = 2A_1B$, then the angle $\widehat{CA_1B}$ is a right angle.